AP PRECALCULUS · TOPIC 1.12

Transformations of Functions

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 The point $(1, 5)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = f(x - 3) + 2$?

A. $(4, 7)$ **B.** $(-2, 7)$

C. $(4, 3)$ **D.** $(-2, 3)$

2 What is the vertex of the graph of $g(x) = -2(x + 4)^2 + 1$, a transformation of $f(x) = x^2$?

A. $(4, 1)$ **B.** $(-4, -1)$

C. $(-4, 1)$ **D.** $(-8, 1)$

3 The point $(6, -3)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = f(2x)$?

A. $(12, -3)$ **B.** $(3, -3)$

C. $(6, -6)$ **D.** $(3, 3)$

4 Let $f(x) = \sqrt{x}$ and $g(x) = f(-x)$. What are the domain and range of g?

A. Domain $[0, \infty)$, range $[0, \infty)$

B. Domain $(-\infty, 0]$, range $(-\infty, 0]$

C. Domain $[0, \infty)$, range $(-\infty, 0]$

D. Domain $(-\infty, 0]$, range $[0, \infty)$

5 The graph of g is the graph of f shifted left 2 units, dilated vertically by a factor of 3, and reflected over the x-axis. Which equation defines g?

A. $g(x) = -3f(x - 2)$ **B.** $g(x) = -3f(x + 2)$

C. $g(x) = -3f(x) + 2$ **D.** $g(x) = -\frac{1}{3}f(x + 2)$

6 The function f has domain $[-4, 8]$. What is the domain of $g(x) = f\left(\frac{x}{2}\right)$?

A. $[-2, 4]$ **B.** $[-4, 8]$

C. $[-8, 16]$ **D.** $[-8, 8]$

7 The function f has range $[1, 5]$. What is the range of $g(x) = -2f(x) + 3$?

A. $[-10, -2]$ **B.** $[-16, -8]$

C. $[5, 13]$ **D.** $[-7, 1]$

8 The point $(4, 2)$ lies on the graph of f. What is the image of this point on the graph of $g(x) = -f\left(\frac{x}{4}\right) + 1$?

A. $(16, -1)$ **B.** $(1, -1)$

C. $(16, 3)$ **D.** $(16, -3)$

Transformations of Functions

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A jet of water from a fountain follows the path $H(x) = -0.4(x-3)^2 + 4.5$, where x is the horizontal distance in metres from the nozzle and H is the height in metres above the water. The path is a transformation of $f(x) = x^2$.

(a) Describe, in order, the transformations of $f(x) = x^2$ that give $H(x)$. DOK3 2 pt

(b) Find the greatest height of the jet and the horizontal distance at which it occurs. DOK3 2 pt

(c) Find the horizontal distance at which the jet returns to the water, to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

The function f has domain $[-3, 5]$ and range $[-2, 6]$, and the point $(2, 4)$ lies on its graph. Let $g(x) = 2f(x-1) + 3$ and $h(x) = -f(2x)$.

(a) Find the domain and the range of g. DOK3 2 pt

(b) Find the image of the point $(2, 4)$ on the graph of g. DOK3 2 pt

(c) Find the domain and the range of h. DOK3 2 pt